

AP CSP | Chapter 1: Ticket Total

Part A | Define, design and build

Name: _____ Partner: _____ Date: _____

Requirements: Valid input is an integer ticket count from 1 to 50. Each ticket costs 12. Add a fee of 3 once per order. Display the total. Invalid-input validation is outside this chapter.

1. Purpose and users: What problem does your program solve, and who would use it?

2. Identify one input, the processing needed, and one output.

3. Write your own pseudocode. Use INPUT(), variables, arithmetic and DISPLAY().

Pair roles: Driver _____ Navigator _____ Roles switched at _____

Credit any code or ideas you adapted, or state that you wrote the solution independently.

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Part B | Test, explain and improve

4. Calculate expected outputs before running. Reset the program between inputs.

Input	Expected output	Actual output	Pass / fail
1			
2			
50			

5. Why are both boundary tests and a typical input useful? Can these tests prove correctness for every possible input?

6. A classmate uses $\text{total} \leftarrow \text{tickets} * (\text{price} + \text{fee})$. Explain why input 1 hides the error and choose an input that reveals it.

7. Feedback and revision: Record a specific suggestion, what you changed, and the result when you retested.

8. Reflection: Name one limitation of your current program and one next improvement.
